

Conditional point processes: Applications to passive market impact

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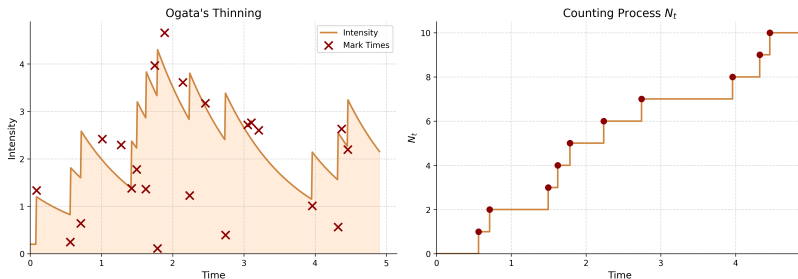
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We observe a point process, how much information about the underlying world does it encode?
How can we deal with the propagation of small shocks to the process?

Thinning for counting processes

Draw random marks in $\mathbb{R}_+ \times \mathbb{R}_+$, describing a random Poisson measure $\pi = \sum_n \delta_{(T_n, Z_n)}$:

$$\lambda_t = f((N_s)_{s < t}, t), \quad N_t = \sum_{\substack{T_n < t \\ Z_n \leq \lambda_{T_n}}} 1$$



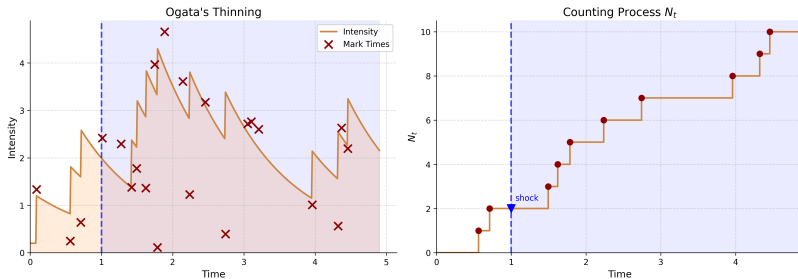
The underlying Poisson measure used for simulation will be called the shadow of the counting process.

Shocking a counting process

We introduce a perturbation of the counting process: adding a deterministic process N^0 to N . The resulting process $\bar{N}$ is now defined as follows:

$$\bar{\lambda}_t = f((N_s)_{s \leq t}, t), \quad \bar{N}_t = \sum_{\substack{T_n \leq t \\ Z_n \leq \bar{\lambda}_{T_n}}} 1 + N_t^0$$

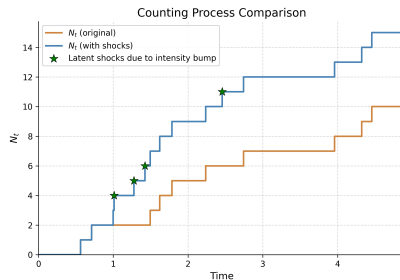
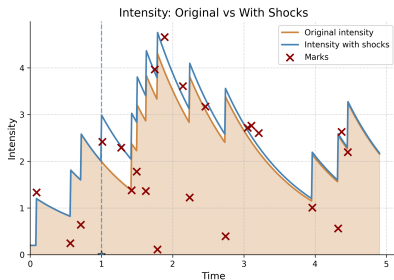
How does this change the process? How does it differ from the original N ?



Shocking with fixed shadow

$$\bar{\lambda}_t = f((N_s)_{s < t}, t), \quad \bar{N}_t = \sum_{\substack{T_n \leq t \\ Z_n \leq \bar{\lambda}_{T_n}}} 1 + N_t^0$$

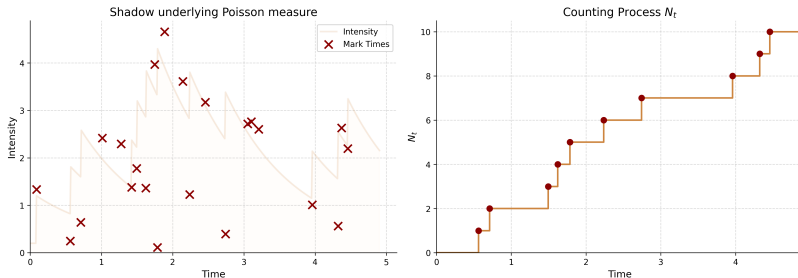
After the shock time, the intensities diverge and jumps propagate while the intensity shock persists.



Understanding the shadow

When observing a counting process, we need to be able to sample from the underlying shadow Poisson measure to understand how shocks could propagate.

In applications, one can only observe N or $\bar{N}$, never the two together.

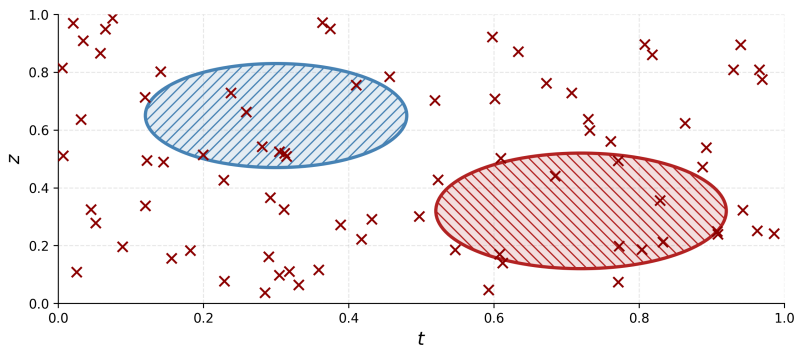


Poisson measures and independence

A Poisson measure π on $\mathbb{R}_+^2$ is a random counting measure such that:

$$\forall (B_i)_{i=1}^p \in \mathcal{B}(\mathbb{R}_+^2)^p \text{ disjoint, } (\pi(B_i))_{i=1}^p \text{ are independent.}$$

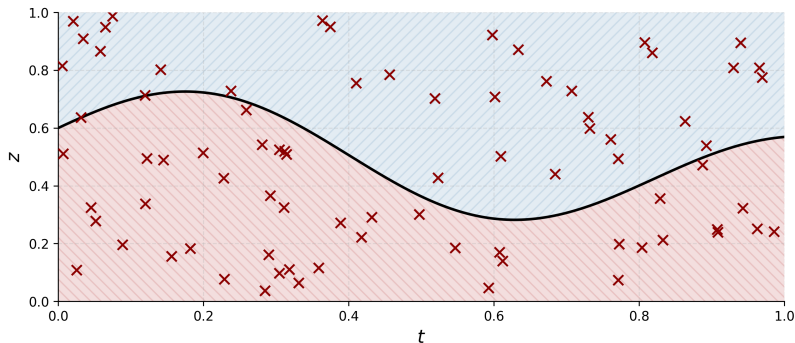
$$\forall B \in \mathcal{B}(\mathbb{R}_+^2), \pi(B) \sim \mathcal{P}(\lambda(B)).$$



The restriction to disjoint sets are independent random sets of points.

Conditional scattering?

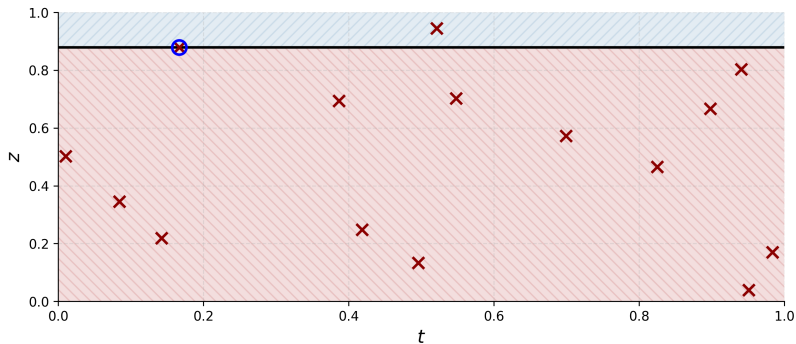
For a Poisson measure π and a random set $A(\pi)$, in what sense can $\pi|_{A(\pi)}$ and $\pi|_{A(\pi)^c}$ be independent?



We want to see if conditionally on the random set $A(\pi)$, they become independent.

Locality through an example

For π in $[0, 1]^2$ and $A(\pi)$ delimiting the horizontal line equal to the second highest mark in π , independence breaks: $\mathbb{P}([0, 1] \times [z, 1] \cap \pi|_{A(\pi)} \neq \emptyset) \cap ([0, 1] \times [z, 1] \cap \pi|_{A(\pi)^c} = \emptyset) | A(\pi)) = 0$



The crucial property is that the random set sitting in some subset S is entirely determined by the restriction $\pi|_S$.

Definition

A random set $A(\pi)$ is called a stopping set if for all compact $K \subset \mathbb{R}_+^2$, $(A(\pi) \subset K)$ is $\pi|_K$ -measurable.

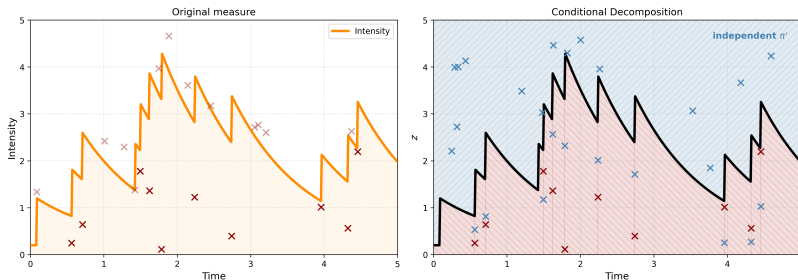
Theorem

For a stopping set $A(\pi)$, $\pi|_{A(\pi)}$ and $\pi|_{A(\pi)^c}$ are independent conditionally on $A(\pi)$. Moreover, $\pi|_{A(\pi)^c} | A(\pi)$ has the law of an independent Poisson measure π' restricted to $A(\pi)$.

Shadowing the counting process

Back to our original problem, we are conditioning on the intensity curve.

- Above the curve: We can now draw points using an independent Poisson measure.
- Under the curve: The mark at a jump time T_m of the counting process is drawn as $Z_m \sim \mathcal{U}([0, \lambda_{T_m}])$.

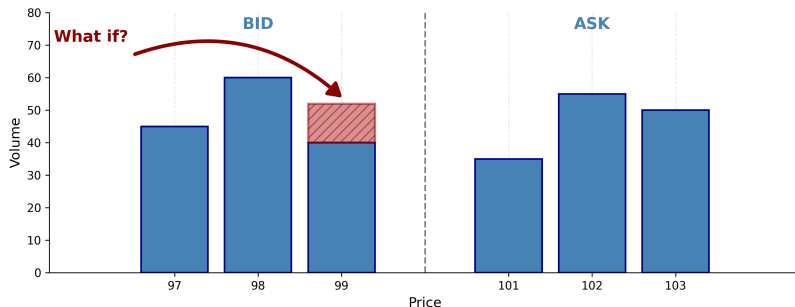


Order book dynamics

Queues of an order book break down to three atomic events:

- Insertion of limit orders: adding an order to the pile.
- Cancellation of orders: removing your own order from the pile.
- Market orders: Consumption of an order present on the pile.

What is the effect of the insertion of a limit order on the price and on the order book?



Restricting our attention to one queue, $q_t := L_t - C_t - N_t$ with L_t, C_t, N_t counting limit, cancel and market orders.

We model L, C as inhomogeneous Poisson with intensities $\lambda^L(q_{t-}), \lambda^C(q_{t-})$, while N will be an independent Hawkes process.